ELEG 487V/FINN 410V
Social, Economic, and Computer Networks

Course Syllabus

General Information:	Instructors:	Dr. Jingxian Wu	Dr. Kathy Fogel
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Textbooks & References:

- Hanneman, Robert A. and Mark Riddle. 2005. Introduction to social network methods. Riverside, CA: University of California, Riverside (published in digital form at http://faculty.ucr.edu/~hanneman/)
- Matthew O. Jackson, 2010, Social and Economic Networks, Princeton University Press.
- David Easley and Jon Kleinberg, 2010, Networks, Crowds, and Markets: Reasoning about a Highly Connected World, Cambridge University Press
- Christakis, Nicholas and James Fowler, 2009. Connected: The Surprising Power of Our Social Networks and How They Shape Our Lives. Little, Brown and Company.
- A. Tannenbaum, 2003, Computer Networks, 4th ed., Prentice Hall.
- W. Stallings, 2004, Data and Computer Communications, 7th ed., Prentice Hall.
- Software: Pajek, Matlab with Boost Graph Library.

Prerequisites: Calculus I & II, Physics I & II

- Knowledge of integration and differentiation
- Knowledge of algebra
- Knowledge of Physics

Learning Objectives: Network theories, basic graph theory, network analysis tools, theories, operations and applications of computer networks, social networks, business networks, and economic networks, applications of network application, entrepreneurship in network applications.

Grading:

- Midterm 25%
- Final 25%
- Homework 25%
- Labs and projects 25%

- A: $90 \leq$ grade ≤ 100
- B: $80 \leq$ grade < 90
- C: $70 \leq$ grade < 80
- D: $60 \leq$ grade < 70
- F: $0 \leq$ grade < 60

- Each student has 1 personal day: you can miss 1 class without losing point.
- Each missing class will result in a 1 point deduction of your final grade.
- Due dates for homework will be strictly enforced. Late submission within one week after due date will receive a 20% grade deduction, and no credit if submitted after one week from the due date.
- If for some legitimate reason (sickness, death in the family, etc.), you cannot take a **test** on the scheduled day, you must notify the instructor **prior** to the exam.

Online Resources:

- Course materials (Slides, Homework, Labs, References, etc) can be found at http://comp.uark.edu/~wuj/teaching/networks/networks.html
- Please check course website **at least once per week** for updates.

Academic Honesty:

Each University of Arkansas student is required to be familiar with and abide by the University's 'Academic Integrity Policy' which may be found at http://provost.uark.edu/ Students with questions about how these policies apply to a particular course or assignment should immediately contact their instructor.

Course Outline:

Chapter 1: Introduction to Networks

1. Definition of networks:
2. Introduction to computer networks
3. Introduction to social networks
4. Introduction to economic networks
5. A unified view of the three networks: what do they share in common, and what are the differences?

Chapter 2: Network Theories

1. Graph representation of networks
2. Basic graph theory
3. Matrix representation of networks
4. Network with special structures: tree, star, circles, and complete networks
5. Properties of networks
6. Network analysis methods

Chapter 3: Network Analysis Tools

1. Overview of network analysis tools
2. Network visualization with Pajek
3. Introduction to Matlab
4. Network analysis with MatlabBGL

Chapter 4: Introduction to Computer Networks

1. Definition: what is computer network?
2. Basics of communications
3. Network classifications
4. Network protocols
5. Protocol layers
6. Network standards
7. Case studies

Chapter 5: Introduction to Social Networks

1. Examples of social networks
2. Cooperation as a survival trait
3. Social network formation
4. Political connection and economic success
5. Using social network analysis to address public health issues
6. The network analysis of crime
7. Case studies

Chapter 6: Introduction to Business and Economic Networks

1. Overview of business and economic networks
2. Banking network, stability, and contagion
3. Herding behavior

4. Why connections matter in business
5. Transportation network
6. Network in Marketing
7. Case studies

Chapter 7: Network Applications in Business

1. Social media: costs and benefits
2. Understanding manager-client relationship using network tools
3. The social network of U.S. CEOs and executives
4. Case studies

Chapter 8: Entrepreneurship in Network Applications (course project)

Labs & Projects:

1. Introduction to network analysis tools
2. Network visualization with Pajek
3. Network analysis with Matlab BGL
4. Shortest path routing of Internet
5. Social network visualization and analysis
6. Business network visualization and analysis
7. Final Project: Entrepreneurship in network applications

Tentative Schedule:

1. Week 1: Ch.1 Introduction to Networks (Wu & Fogel)
2. Week 2: Ch. 2 Network Theories (Sections 1, 2) (Wu)
3. Week 3: Ch. 2 Network Theories (Sections 3-5) (Wu)
4. Week 4: Ch. 2 (Section 6); Ch. 3 Analysis Tools (Sections 1, 2) (Wu)
5. Week 5: Ch. 3 Analysis Tools (Sections 2-4) (Wu)
6. Week 6: Ch. 4 Computer Networks (Sections 1-3) (Wu)
7. Week 7: Ch. 4 Computer Networks (Section 4), Midterm (Wu)
8. Week 8: Ch. 4 Computer Networks (Section 5-7) (Wu)
9. Week 9: Ch. 5 Social Networks (Sections 1-4) (Fogel)
10. Week 10: Ch. 5 Social Networks (Sections 5-7) (Fogel)
11. Week 11: Ch. 6 Business and Economic Networks (Sections 1-3) (Fogel)
12. Week 12: Ch. 6 Business and Economic Networks (Sections 4-6) (Fogel)
13. Week 13: Ch. 7 Network Applications in Business (Fogel)
14. Week 14: Ch. 8 Entrepreneurship in Network Applications (Fogel)
15. Week 15: Ch. 8 Entrepreneurship in Network Applications (Fogel & Wu)

The above schedule is subject to change without prior notice.